

Recovering the partial-solution strata by membership alone: a GCRD closure construction

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A companion note to *A New Method of Solving PDEs*, and a sequel to the note *Membership versus variety, and the partial-solution strata*. That note established that the core (membership) algorithm computes the locus $\mathcal{V} = \{c : P \in [A(c)]\}$ — a *universal* condition — and is therefore blind, by construction, to the *partial-solution strata*: parameter values c at which a proper nontrivial subfamily of the ansatz solutions satisfies the PDE while the family as a whole does not. Capturing those strata seemed to require adjoining P to the ansatz and running a full differential decomposition — the expensive existential computation.

This note shows that, for ansätze whose top element is a *linear* ODE, the existential computation is unnecessary. Given such an ansatz A , we construct — by pure polynomial linear algebra, with no differential elimination beyond the Ritt reduction the core algorithm already performs — a finite set of auxiliary ansätze $A_1, \dots, A_{n-1}$ of the same class and lower order, such that every partial-solution stratum of A is an ordinary *membership* stratum of some A_k . Running the core algorithm on the collection then achieves the completeness of the existential formulation at membership-formulation cost. The engine is the greatest common right divisor (GCRD) of Ore polynomials, made parametric through the subresultant theory of Chardin and Li [2, 3]; as a by-product, the partial strata themselves acquire explicit polynomial defining equations in the *original* constant space.

1 The covering specification

Throughout, $A(c)$ is an ansatz with constant parameters $c \in \mathbb{C}^{n_c}$, P is the PDE, and $\mathcal{V}(A) = \{c : P \in [A(c)]\}$ is the membership locus computed by the core algorithm (Ritt reduction of P modulo A , projection of the remainder onto the constants).

Definition 1 ($\forall$ -cover). *A finite collection of ansätze $\{A_1, \dots, A_m\}$ (each with its own constants) $\forall$ -covers A with respect to P if, for every c^* and every nonzero $\Psi_0 \in \text{Sol}(A(c^*)) \cap \text{Sol}(P)$, there exist an index i and constants c' with*

$$\Psi_0 \in \text{Sol}(A_i(c')) \quad \text{and} \quad c' \in \mathcal{V}(A_i).$$

If a collection $\forall$ -covers A , then running the core algorithm on each member delivers every solution function that the existential computation on A would deliver — each one packaged inside a family that solves P wholesale, which is the form of answer the method is designed to produce. Two requirements make the notion useful rather than vacuous:

1. The collection must be constructed from A and the *syntactic shape* of P (its order and degree bounds) alone. If the construction required decomposing $A \cup \{P\}$, it would cost what it saves.
2. The collection must be finite, with all degree bounds computable.

2 Setting: a single linear ODE element

We treat the shape of the paper's Figure "ansatz 5": a base differential ring B (e.g. $\mathbb{Q}\{x, y, z\}[r]/(r^2 - x^2 - y^2 - z^2)$), a linear form $v = v_1x + v_2y + v_3z + v_4r$ with constant coefficients, the chain-rule relations $\Psi_x = \Psi'v_x$, etc., and a top element

$$L = \sum_{i=0}^n a_i(v) \partial^i, \quad a_i \in K[v], \deg_v a_i \leq d, a_n \neq 0, \quad (1)$$

regarded as an Ore (differential) operator acting on Ψ as a function of v ; here $K = \mathbb{Q}(c)$ and $\partial = d/dv$. The natural home for the operator computations is the right-Euclidean ring $K(v)[\partial]$, in which every left ideal is principal and greatest common right divisors exist. Two standard facts [1, 4] are used repeatedly:

1. (G exists and is rational.) For operators $L, R_1, \dots, R_s \in K(v)[\partial]$, the left ideal $K(v)[\partial]L + \sum_{\beta} K(v)[\partial]R_{\beta}$ is principal; its monic generator $G = \text{GCRD}(L, R_1, \dots, R_s)$ right-divides each argument and is computable by the right Euclidean algorithm, with coefficients in $K(v)$.
2. (Kernels intersect.) On a domain where the coefficients are analytic and $a_n \neq 0$, the local solution space of an order- m operator is m -dimensional, and

$$\ker G = \ker L \cap \bigcap_{\beta} \ker R_{\beta}.$$

The remainder as a system of operators. Let P be a differential polynomial *linear* in Ψ and its derivatives. Ritt's full reduction of P modulo A yields $hP - r \in [A]$ with r fully reduced; since both P and the elements of A are linear in the Ψ -jets, so is r , and full reducedness confines it to $\Psi, \Psi', \dots, \Psi^{(n-1)}$. Its coefficients lie in the base ring extended by v . Expanding along a set of power products m_{β} in the base indeterminates that remain linearly independent over $K(v)$ on the solution locus (Remark 1), we write

$$r = \sum_{\beta} m_{\beta} \cdot R_{\beta}(\Psi), \quad R_{\beta} \in K(v)[\partial], \quad \text{ord } R_{\beta} \leq n-1. \quad (2)$$

In the univariate case (base ring $\mathbb{Q}\{x\}$, $v = x$) there is a single β and $R_1 = r$ itself.

Remark 1 (base-splitting hypothesis). *Because v is itself a linear form in the base variables, monomials in x, y, z, r are not automatically independent over $K(v)$: the relation $v_1x + v_2y + v_3z + v_4r - v = 0$ is a dependence. A valid splitting eliminates one base coordinate in favor of v (possible whenever the form v is nondegenerate, a genericity condition on the v_i of the same character as the paper's hypothesis (2)) and expands r in the standard monomials of the remaining coordinates. We assume such a splitting has been fixed; all statements below are relative to it.*

Lemma 1 (solution criterion). *Let c^* satisfy $h(c^*) \neq 0$. For every solution Ψ_0 of $A(c^*)$:*

$$P(\Psi_0) = 0 \iff R_{\beta}(c^*) \Psi_0 = 0 \text{ for all } \beta.$$

Proof. ($\Leftarrow$) All $R_{\beta}(c^*) \Psi_0 = 0$ gives $r(c^*)(\Psi_0) = 0$ by (2); the reduction identity gives $h(c^*) P(\Psi_0) = r(c^*)(\Psi_0) = 0$, and $h(c^*) \neq 0$. ($\Rightarrow$) $P(\Psi_0) = 0$ forces $r(c^*)(\Psi_0) \equiv 0$ on the domain; by the independence of the m_{β} over functions of v (Remark 1), each coefficient function $R_{\beta}(c^*) \Psi_0$ vanishes. $\square$

3 The GCRD structure theorem

Fix c^* with $h(c^*) \neq 0$ and set

$$G(c^*) = \text{GCRD}(L(c^*), R_1(c^*), \dots, R_s(c^*)) \in \mathbb{C}(v)[\partial], \quad k = \text{ord } G(c^*).$$

Since G right-divides L , we have $0 \leq k \leq n$. Write $\tilde{G}$ for G with denominators cleared (polynomial coefficients, content removed), and let $A_G(c^*)$ denote the ansatz $A(c^*)$ with the top element $L(c^*)$ replaced by $\tilde{G}(c^*)$ — the same tower, one order lower.

Theorem 1 (trichotomy and refinement). *With the hypotheses of Lemma 1:*

1. $k = n$ iff every $R_\beta(c^*) = 0$, iff $c^* \in \mathcal{V}(A)$ (membership: the whole family solves P).
2. $k = 0$ iff $\text{Sol}(A(c^*)) \cap \text{Sol}(P) = \{0\}$ (no stratum).
3. $0 < k < n$ iff c^* lies on a partial-solution stratum, and then

$$\text{Sol}(A(c^*)) \cap \text{Sol}(P) = \text{Sol}(A_G(c^*)),$$

a k -dimensional family: exactly the solutions of the order- k operator $G(c^)$, transported through the unchanged lower tower.*

Proof. By Lemma 1 and fact (2) of §2, $\text{Sol}(A(c^*)) \cap \text{Sol}(P)$ consists of the solutions of the tower whose top function lies in $\ker L \cap \bigcap_\beta \ker R_\beta = \ker G$. If every $R_\beta(c^*) = 0$ the left ideal is generated by L alone, $G \sim L$ and $k = n$; the projection coefficients of r vanish at c^* , which is the definition of $\mathcal{V}(A)$. If $k = 0$ then G is a unit and the common kernel is trivial. Otherwise $\ker G$ is k -dimensional with $0 < k < n$: a proper nontrivial subfamily, i.e. a partial stratum in the sense of the previous note's Definition 1; and since G right-divides $L(c^*)$, every solution of $A_G(c^*)$ solves $A(c^*)$, giving the displayed equality. $\square$

The next statement is the reason the construction pays: the refined system is not merely consistent with P — it *members* P , so the core algorithm, run on a generic family containing A_G , finds it.

Theorem 2 (membership transfers to the refinement). *Let $0 < k < n$ as in Theorem 1(3), and let A_k be the generic ansatz obtained from A by replacing the top element with an undetermined linear ODE of order k ,*

$$L_k = \sum_{j=0}^k b_j(v) \partial^j, \quad \deg_v b_j \leq B_k,$$

whose coefficients' v -coefficients form the new constant vector c' (the bound B_k is fixed in Proposition 2). Let $c'(c^)$ be the coefficient vector of $\tilde{G}(c^*)$. Then*

$$c'(c^*) \in \mathcal{V}(A_k),$$

i.e. the Ritt remainder of P modulo A_k , projected onto the constants, vanishes at $c'(c^)$.*

Proof. Reduce P modulo A_k over $K(c')$: $h_k P - r_k \in [A_k]$ with r_k linear in $\Psi, \dots, \Psi^{(k-1)}$, and split $r_k = \sum_\beta m_\beta R_{k,\beta}$ as in (2), so each $R_{k,\beta} \in K(c')(v)[\partial]$ has order $\leq k-1$. Specialize $c' = c'(c^*)$; note $h_k(c'(c^*)) \neq 0$, since the initial and separant of the monic-in- $\Psi^{(k)}$ element $\tilde{G}$ reduce to its leading v -coefficient, a nonzero polynomial. Every $\Psi_0 \in \text{Sol}(A_k(c'(c^*))) = \text{Sol}(A_G(c^*))$

solves P by Theorem 1(3), so Lemma 1, applied to A_k , gives $R_{k,\beta}(c'(c^*))\Psi_0 = 0$ for all β and all Ψ_0 in a k -dimensional space. An operator of order $\leq k-1$ annihilating k linearly independent functions (nonzero Wronskian) is the zero operator. Hence every $R_{k,\beta}(c'(c^*)) = 0$; equivalently, every projection coefficient of r_k vanishes at $c'(c^*)$, which is $c'(c^*) \in \mathcal{V}(A_k)$. $\square$

Note the Wronskian argument replaces the characteristic-set machinery of the paper's Theorem 3 entirely — linearity is doing all the work.

4 Parametric stratification and the degree bound

Theorems 1 and 2 are pointwise in c^* . The subresultant theory for Ore polynomials [2, 3] makes them parametric. For a pair L, R of orders $n > n'$, the k -th subresultant $S_k(L, R)$, $0 \leq k \leq n'$, is an operator of order k whose coefficients are determinants of Sylvester-type matrices of size $n + n' - 2k$ built from the coefficients of $\partial^j R$ ($0 \leq j < n - k$) and $\partial^j L$ ($0 \leq j < n' - k$); as in the commutative theory,

$$\text{ord GCRD}(L, R) = \min\{k : S_k \not\equiv 0\}, \quad \text{GCRD}(L, R) \sim S_{k_{\min}}.$$

Since Ritt's full reduction is fraction-free, the coefficients of L and of every R_β are polynomials in c and v ; hence so is every entry of every subresultant matrix, and we obtain:

Proposition 1 (strata equations). *For each k , the locus $\{c : \text{ord } G(c) \geq k\}$ is closed, cut out by the vanishing of the v -coefficients of the subresultants $S_0, \dots, S_{k-1}$ — explicitly computable polynomials in c . In particular the union of the partial-solution strata is the constructible set $\{\text{ord } G \geq 1\} \setminus \mathcal{V}(A)$, now presented by polynomial equations and inequations in the original constants, refining the inclusion $\pi_c(\text{Sol}(A) \cap \text{Sol}(P)) \supseteq \mathcal{V}$ of the previous note into a computable stratification. Moreover, on the stratum $\{\text{ord } G = k\}$ the coefficients of $\tilde{G}$ — hence the transfer map $c \mapsto c'(c)$ of Theorem 2 — are polynomial in c .*

Proposition 2 (degree bound). *Let D bound the v -degrees of the coefficients of L and of all R_β , and $n' = \max_\beta \text{ord } R_\beta \leq n - 1$. Then the v -degrees of the coefficients of S_k — and hence of $\tilde{G}$ on any stratum where $\text{ord } G = k$ — are at most*

$$B_k = (n + n' - 2k)D.$$

Proof sketch. Left-multiplying by ∂ maps an operator with polynomial coefficients of v -degree $\leq D$ to one with coefficient degrees $\leq D$ (differentiation does not raise polynomial degree). Every entry of the k -th matrix is a coefficient of some $\partial^j L$ or $\partial^j R_\beta$ and so has degree $\leq D$; the determinant of a square matrix of size $n + n' - 2k$ then has degree $\leq (n + n' - 2k)D$. $\square$

The bound is coarse but uniform in c , which is what the construction needs. (For $s > 1$ operators R_β , compute the GCRD by pairwise iteration, $\text{GCRD}(L, R_1, R_2) = \text{GCRD}(\text{GCRD}(L, R_1), R_2)$, clearing denominators at each step; the bounds compose and remain computable, at the cost of an explicit closed formula.)

5 The algorithm

Input: an ansatz A of the shape of §2 (top element linear of order n , coefficient degrees $\leq d$); the order and degree shape of the PDE P .

Output: ansätze $A_1, \dots, A_{n-1}$ such that $\{A, A_1, \dots, A_{n-1}\}$ $\forall$ -covers A off the bad locus; optionally, the partial strata as subvarieties of the original c -space.

1. **Reduce.** Ritt-reduce P modulo A over $\mathbb{Q}(c)$ (this is the core algorithm's own first step), obtaining h and r ; split r over base power products as in (2) to get the operators R_β .
2. **Stratify.** For $k = 1, \dots, n-1$ compute the subresultant data of L against $\{R_\beta\}$: the strata polynomials of Proposition 1 and the degree bounds B_k of Proposition 2.
3. **Emit.** For each k , output A_k : the tower of A with the top element replaced by a generic linear ODE of order k with polynomial coefficients of v -degree $\leq B_k$, its coefficient list forming the new constants c' . Record the transfer maps $c \mapsto c'(c)$ (coefficients of $\tilde{G}$ on each stratum).
4. **Run.** Apply the core algorithm — projection plus the bad-locus recursion of the paper's general algorithm — to A and to each A_k ; return the union of the membership loci (and, if the original-space geometry is wanted, the strata of step 2).

Theorem 3 (covering). *Let A be as in §2 and P linear of the given shape. For every c^* with $h(c^*) \neq 0$ (and satisfying the base-splitting genericity of Remark 1) and every nonzero $\Psi_0 \in \text{Sol}(A(c^*)) \cap \text{Sol}(P)$: either $c^* \in \mathcal{V}(A)$, or, with $k = \text{ord } G(c^*) \in \{1, \dots, n-1\}$,*

$$\Psi_0 \in \text{Sol}(A_k(c'(c^*))) \quad \text{and} \quad c'(c^*) \in \mathcal{V}(A_k).$$

Hence $\{A, A_1, \dots, A_{n-1}\}$ $\forall$ -covers A away from the bad locus $\{h = 0\}$; parameter values on the bad locus are handled, exactly as in the paper, by the general algorithm's recursion, with the present construction applied to each ansatz that recursion produces (both recursions strictly decrease their measures — order here, the bad-locus dimension there — so the composition terminates).

Proof. Immediate from Theorems 1 and 2, Proposition 2 guaranteeing that $\tilde{G}$'s coefficients fit within A_k 's degree budget. $\square$

6 The two-line example, closed

The previous note's Example 1 exhibited the gap; the construction now closes it mechanically. Take

$$A(c) : \Psi'' - c\Psi = 0 \quad P : \Psi' - \Psi = 0,$$

univariate base, so $L = \partial^2 - c$ and the remainder is the single operator $R = \partial - 1$ ($n = 2$, $n' = 1$, $D = 0$).

Step 1–2. One right division,

$$\partial^2 - c = (\partial + 1)(\partial - 1) + (1 - c),$$

shows the 0-th subresultant (the differential resultant) is $1 - c$ up to a unit. Strata: for $c \neq 1$, $G = 1$ — no common solutions; on $\{c = 1\}$, $G = \partial - 1$, order $k = 1$. The partial stratum of the previous note is recovered as the vanishing locus of one polynomial in c , with no decomposition performed.

Step 3. $B_1 = (n + n' - 2k)D = 0$: emit $A_1(c') : \Psi' - c'\Psi = 0$, constant coefficient. Transfer map: $c'(c) = 1$ on the stratum.

Step 4. The core algorithm on A_1 reduces P to $r_1 = (c' - 1)\Psi$, so $\mathcal{V}(A_1) = \{c' = 1\}$ — the family $\Psi = Ke^x$, containing the witness that was invisible to $\mathcal{V}(A) = \emptyset$. The previous note's §4 family, introduced there by hand, is here *derived*, together with the stratum equation $c = 1$ and the point $c' = 1$ it maps to.

7 Beyond a single linear element

Towers (nested ODE elements). For ansätze like the paper’s Figure “ansatz 12”, a witness may instead constrain the integration constants of a *lower* level (a special Θ). The construction recurses: the subresultant coefficients from step 2, which now live in the lower tower’s differential field rather than in $K(v)$, become the constraint one level down; treat that level as top and repeat. GCRD theory is valid over any differential field, so Theorems 1 and 2 survive level by level; what must be re-proved per level is the *uniform degree bound* (the analogue of Proposition 2 over the lower tower’s fraction field). That lemma is of the same character as the bounded-prolongation note’s N_0 bound and is left open here. Orders strictly decrease and the tower is finite, so the recursion terminates in a finite collection regardless.

Nonlinear first-order elements. The vector-space machinery is linear to its bones, but the first-order nonlinear case has its own closure. Full reduction against a first-order element leaves a remainder containing *no* Ψ' at all, so a partial-stratum witness satisfies the *algebraic* relation $r(v, \Psi) = 0$; the refined system replaces the ODE element by an irreducible factor q of r as an *algebraic extension element* (a “green box”), of degree $\leq \deg_{\Psi} r$ — a bound available from the reduction. Compatibility of q with the derivation becomes polynomial conditions on the constants, playing the role of the transfer map. So the closure of a Riccati-type element consists of algebraic-extension ansätze of bounded degree, again entirely within the paper’s formalism.

Higher-order nonlinear elements. Open. Witnesses may sit on singular components (separant loci), where neither the GCRD argument nor the algebraic-factor argument applies; Ritt’s theory of singular solutions is the natural starting point, and no bounded-shape theorem is claimed.

8 Reading

The membership algorithm’s blindness to the partial-solution strata is not a price of its cheapness — for linear ODE ansätze it is a bookkeeping gap, closed by enlarging the ansatz *collection* rather than the *algorithm*. Everything added is linear algebra over $K(v)$ with entries polynomial in the constants: no differential elimination is performed beyond the single Ritt reduction the core algorithm already does. The existential formulation’s extra content — which subvarieties of constant space carry partial solutions, and what those solutions are — reappears as the subresultant stratification (Proposition 1) and the lower-order membership families (Theorem 3).

The case $k = 1$ connects to classical territory: a one-dimensional common kernel is an exponential solution, i.e. a first-order right factor, the object computed by the Beke and van Hoeij factorization algorithms [5, 6]. Their local (Newton-polygon) degree bounds are sharper than Proposition 2’s determinant bound, but are pointwise in c ; the subresultant bound is cruder and uniform, which is what a parametric family construction requires. Where the sharper bounds matter in practice, they can be substituted stratum by stratum.

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